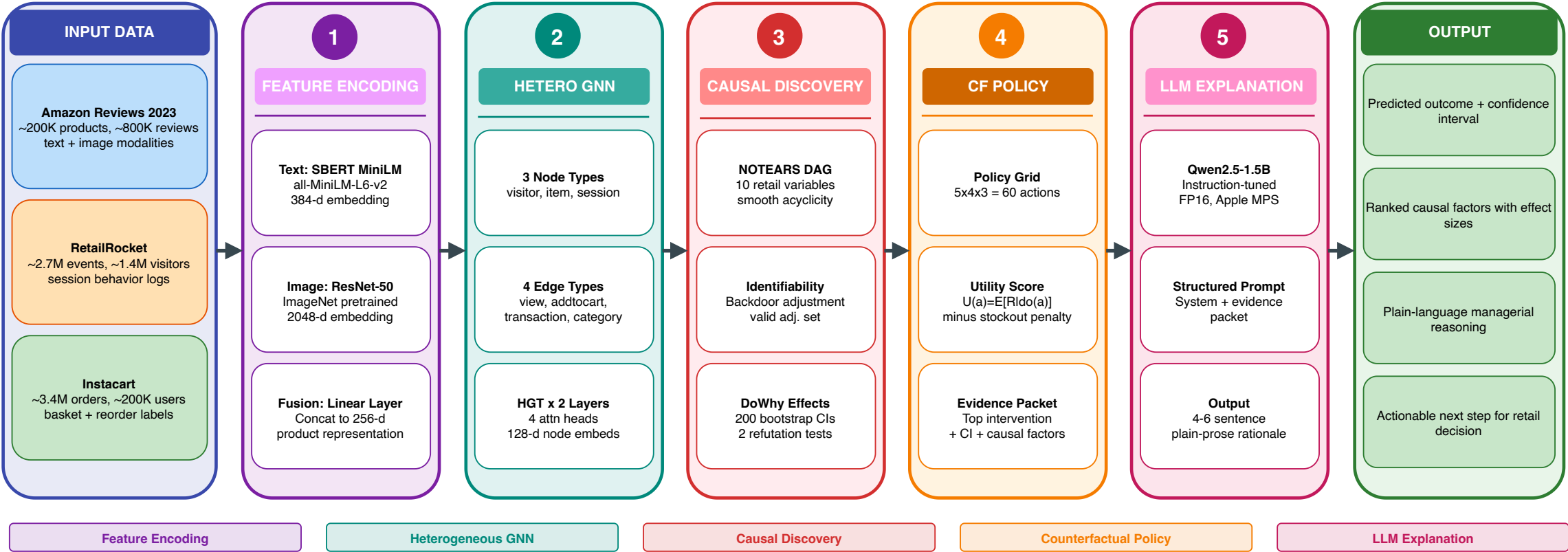
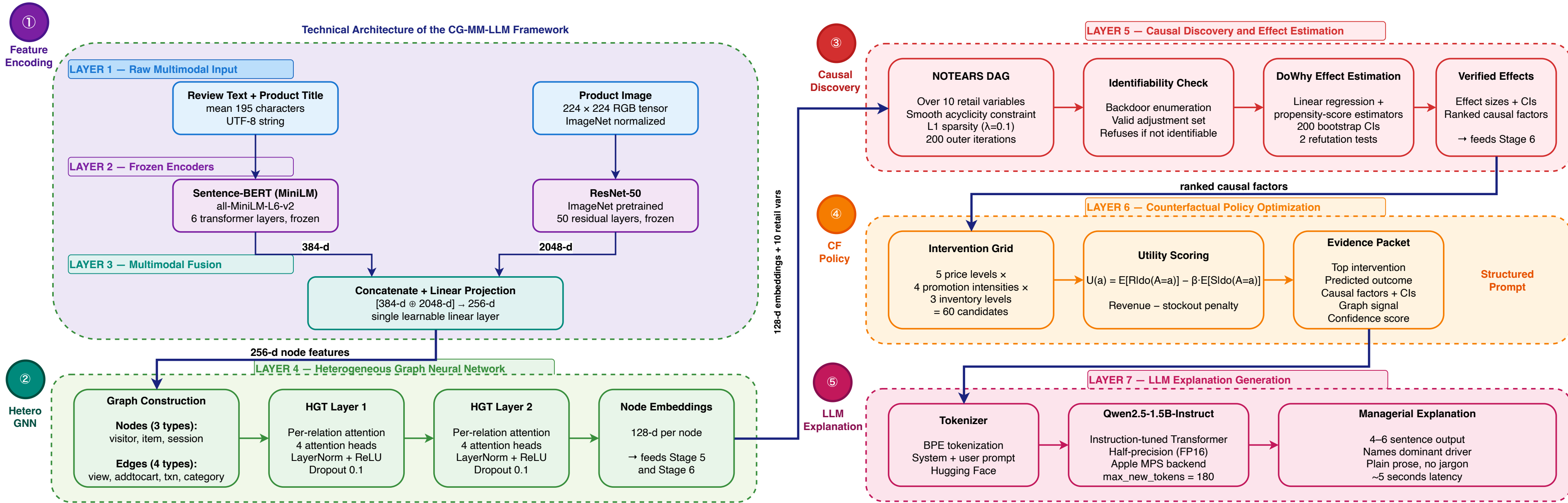


End-to-End Workflow of the CG-MM-LLM Framework

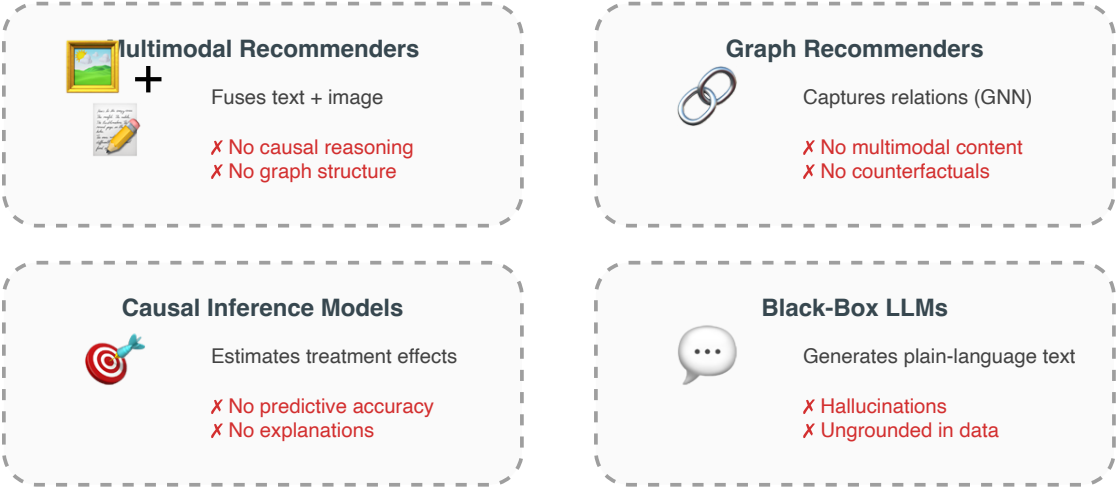
Raw multimodal retail data to causally grounded plain-language decision recommendation





From Fragmented Approaches to Integrated Retail Decision Intelligence

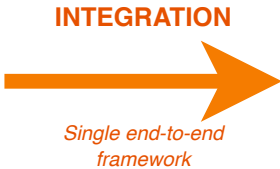
CURRENT STATE — Siloed Approaches



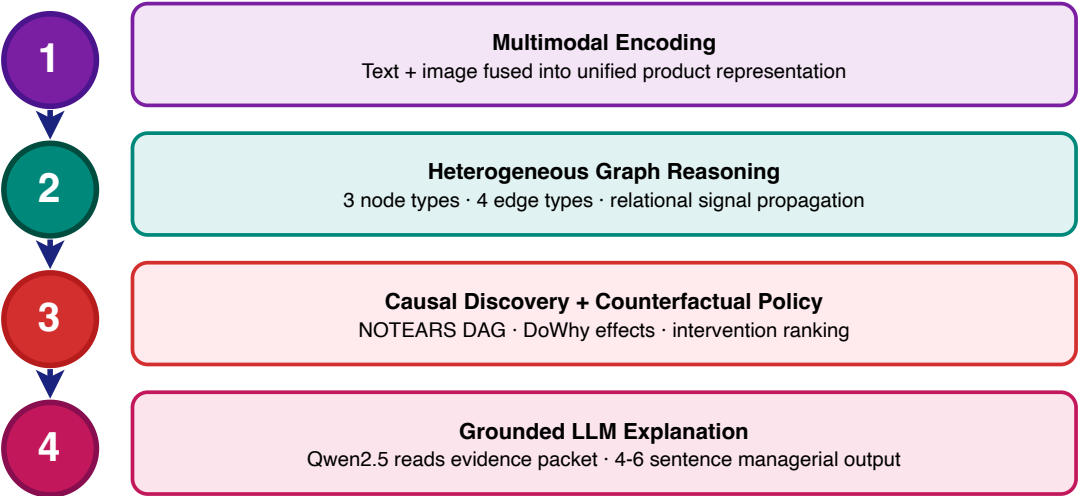
⚠ THE INTEGRATION GAP

Each approach solves only one piece of the retail decision problem.

No prior system has unified all four into a single end-to-end pipeline tested on real public retail data, leaving retail managers without a complete decision-support tool.



PROPOSED FRAMEWORK — Integrated Pipeline



✓ Delivers what no individual approach can



Composite Decision Intelligence Score: DI = 0.707

+27.8 percent over best baseline · 96.4 percent retention across robustness conditions · all four task dimensions improved simultaneously

Evaluated on three real public retail datasets · ~4 million records · commodity Apple Silicon M4 hardware (no cloud GPU required)

 Amazon Reviews 2023

 RetailRocket

 Instacart

Honest Contribution: gains in 3 of 4 task dimensions, with limitations transparently documented